

Nick's Question 0.1 Details from Remark 1.1 page 3 that part (i) implies the existence of K .

Is this correct in showing

Let $\psi \in C_0^\infty(\mathbb{R}^d)$ then we have

$$\begin{aligned} \int_{\mathbb{R}^d} dx \gamma(x) \psi(x) &= \int_{\mathbb{R}^d} dx \varphi(x) \mathcal{F}(\psi)(x) \\ &= \int_{\mathbb{R}^d} dx \mathcal{F}(\varphi)(x) \psi(x) \\ &= \int_{\mathbb{R}^d} dx \mathcal{F}(\sqrt{\varphi} \sqrt{\varphi})(x) \psi(x) \\ &= \int_{\mathbb{R}^d} dx \left(\mathcal{F}(\sqrt{\varphi}) * \mathcal{F}(\sqrt{\varphi}) \right)(x) \psi(x) \end{aligned}$$

and since this is true for all test functions ψ then we must have $\gamma(x) = \left(\mathcal{F}(\sqrt{\varphi}) * \mathcal{F}(\sqrt{\varphi}) \right)(x)$

Nick's Question 0.2 top of page 6

Why does assumption A imply that $\alpha \in (0, 2 \wedge d)$?

Nick's Question 0.3 page 8

Why do we have the representation (2.3)?

Nick's Question 0.4 Lemma 2.3

1. Why does $f(cA) = c^{d-\alpha} f(A)$ imply that when f is a density then $f(cx) = c^{-\alpha} f(x)$?
2. Does

$$\int_{\mathbb{R}^d} |\xi|^2 |\mathcal{F}g(\xi)|^2 d\xi = \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx?$$

3. Why does $(\tilde{g}^2 * f)(x) = B^{-\alpha} (g^2 * f)(B^{-1}x)$. Here is what I have

$$\begin{aligned} B^{-\alpha} (g^2 * f)(B^{-1}x) &= B^{-\alpha} \int_{\mathbb{R}^d} g^2(y) f\left(\frac{x}{B} - y\right) dy \\ &= \int_{\mathbb{R}^d} g^2(y) f(x - By) dy \end{aligned}$$

since $B^{-\alpha} f\left(\frac{x}{B} - y\right) = f(x - By)$. Then,

$$\begin{aligned} \int_{\mathbb{R}^d} g^2(y) f(x - By) dy &= B^{-d} \int_{\mathbb{R}^d} g^2\left(\frac{u}{B}\right) f(x - u) du \\ &= B^{-d/2} \int_{\mathbb{R}^d} \tilde{g}^2(u) f(x - u) du \\ &= B^{-d/2} (\tilde{g}^2 * f)(x) \end{aligned}$$

and this shows that

$$B^{-d/2} (\tilde{g}^2 * f)(x) = B^{-\alpha} (g^2 * f)(B^{-1}x)$$

4. If I am correct on (3) then

$$\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{d-\alpha} \langle g^2 * f, g^2 \rangle$$

5. If I am incorrect in (3) then how do we use $(\tilde{g}^2 * f)(x) = B^{-\alpha}(g^2 * f)(B^{-1}x)$ to get that $\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{-\alpha} \langle g^2 * f, g^2 \rangle$. We have that

$$\begin{aligned} \langle \tilde{g}^2 * f, \tilde{g}^2 \rangle &= \int_{\mathbb{R}^d} (\tilde{g}^2 * f)(x) \tilde{g}^2(x) dx \\ &= \int_{\mathbb{R}^d} B^{-\alpha}(g^2 * f)(B^{-1}x) \tilde{g}^2(x) dx \\ &= \int_{\mathbb{R}^d} B^{-\alpha}(g^2 * f)(B^{-1}x) B^{-d/2} g^2(B^{-1}x) dx \\ &= B^{-\alpha-\frac{d}{2}} \int_{\mathbb{R}^d} (g^2 * f)(B^{-1}x) g^2(B^{-1}x) dx \\ &= B^{-\alpha+\frac{d}{2}} \int_{\mathbb{R}^d} (g^2 * f)(u) g^2(u) du \\ &= B^{-\alpha+\frac{d}{2}} \langle g^2 * f, g^2 \rangle \end{aligned}$$

and this is also different from the paper.

Nick's Question 0.5 page 11 When $d = 3$ we know that $G(t, \cdot)$ is a measure so does $(G(t, \cdot) * p_\epsilon)(x) = \int p_\epsilon(x - y) G(t, dy)$?

Nick's Question 0.6 page 11 typos

1. on the middle of page you say $d \leq 3$ right before (3.6) but I think it need to be $d = 3$.
2. When defining L and L_ϵ . There is a dx in both integrals but there needs to be a dt .
3. When applying the change of variables to L_n and $L_{n,\epsilon}$ the index on the sum I think should be Σ_n not S_n .

Nick's Question 0.7 How are the equations on the bottom of page 11 proved. For example, using (3.3), we see that

$$L_1(y_1) = \int_0^\infty e^{-t} \int_0^t G(s, y_1) ds dt$$

but according to the equation for L_n on the bottom of page 11 we should also have

$$L_1(y_1) = L(y_1)$$

where

$$L(x) = \int_0^\infty e^{-t} G(t, x) dt$$

But it doesnt seem that $L_1(y_1) = L(y_1)$. For example, consider $d = 1$ then we have that

$G(t, x) = \frac{1}{2} 1_{|x| < t}$ and so

$$\begin{aligned}
L_1(y_1) &= \int_0^\infty e^{-t} \int_0^t G(s, y_1) ds dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} \int_0^t 1_{\{|y_1| < s\}} ds dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} \int_{|y_1| \wedge t}^t ds dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} (t - (|y_1| \wedge t)) dt \\
&= \frac{1}{2} \int_{|y_1|}^\infty e^{-t} (t - |y_1|) dt
\end{aligned}$$

However

$$\begin{aligned}
L(y_1) &= \int_0^\infty e^{-t} G(t, y_1) dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} 1_{\{|y_1| < t\}} dt \\
&= \frac{1}{2} \int_{|y_1|}^\infty e^{-t} dt
\end{aligned}$$

and these do not seem to be equal.

Nick's Question 0.8 Lemma 3.3 What are the details showing

$$\int_{(\mathbb{R}^d)^n} \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) L_n(y_1, \dots, y_n) dy \right]^2 dx = \int_{(\mathbb{R}^d)^n} |\mathcal{F} L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n)$$

for example suppose that $n = 1$ so we want to show

$$\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} K(x - y) L_n(y) dy \right]^2 dx = \int_{\mathbb{R}^d} |\mathcal{F} L_n(\xi)|^2 \mu(d\xi)$$

By Plancherels Theorem we have

$$\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} K(x - y) L_n(y) dy \right]^2 dx = \int_{\mathbb{R}^d} \left[\mathcal{F} \left(\int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy \right) (\xi) \right]^2 d\xi$$

However anyway I continued seemed not to work. I thought we should use the property that $\mathcal{F}(f)\mathcal{F}(g) = \mathcal{F}(f * g)$ and this would give us

$$\int_{\mathbb{R}^d} \left[\mathcal{F} \left(\int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy \right) (\xi) \right]^2 d\xi = \int_{\mathbb{R}^d} \mathcal{F} \left(\int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy * \int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy \right) (\xi) d\xi$$

but continuing from this didnt seem to work.

Nick's Question 0.9 Lemma 3.4

1. Why does $\gamma(y_k - y'_k) = \int_{\mathbb{R}^d} K(y_k - x_k)K(y'_k - x_k)dx_k$. We know that $\gamma = K * K$ so

$$\begin{aligned}\gamma(y_k - y'_k) &= \int_{\mathbb{R}^d} K(y_k - y'_k - x_k)K(x_k)dx_k \\ &= \int_{\mathbb{R}^d} K(y_k - u)K(u - x_k)du\end{aligned}$$

by letting $u = y'_k + x_k$

2. Where are the first and second equalities coming from?

Nick's Question 0.10 Lemma 3.6 page 14
above equation (3.13)?

How do we get this inequality right